Cover letter

Dear Editors,

I submit the manuscript entitled "Mobility Suppression Boundary Principle for Polymer Glass-Transition Temperature" for consideration as an original research article. The manuscript presents an interpretable polymer-informatics analysis of glass-transition temperature boundary displacement using a public-safe reproducibility workflow. The study evaluates whether suppression of normalized local rotatable mobility, expressed as MSBP density, is associated with upward Tg boundary displacement within comparable chemistry-size fibers.

The contribution is not presented as a universal Tg law or as a replacement for experimental measurement. Instead, it provides a falsification-controlled and reproducible boundary-coordinate analysis, supported by source-family checks, bootstrap summaries, paired descriptor comparisons, shuffle controls, contradiction taxonomy, and a public GitHub/Zenodo reproducibility package.

A preprint version is available at ChemRxiv:

https://doi.org/10.26434/chemrxiv.15005629/v1

The public-safe reproducibility package is archived at Zenodo:

https://doi.org/10.5281/zenodo.21100020

The code repository is available at:

https://github.com/htetkokokonaing-dev/msbp-tg

Thank you for considering this manuscript.

Sincerely,

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Independent Researcher

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